

SMART DUSTBIN

WASTE SEGREGATION

Sensor Lab Project Idea

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PROJECT DESCRIPTION

The Smart Dustbin is an innovative system designed to automatically segregate waste into three categories: dry, wet, and metal. Using sensors, the system sorts waste efficiently into separate compartments, promoting better recycling and waste management.

Key Features:

Automated Waste Detection:

Sensors identify and classify waste (dry, wet, or metal).

Waste Segregation:

Automatically directs waste to the correct compartment.

Multiple Compartments:

Separate bins for dry, wet, and metal waste.

Environmentally Friendly:

Promotes recycling and reduces landfill waste.

CRITERIA

- Sensors: IR sensor, Proximity sensor, Rain drop moisture sensor
- 1 Micro - Controller: Arduino uno smd
- WiFi Tech: Bluetooth

UNIQUE FEATURE

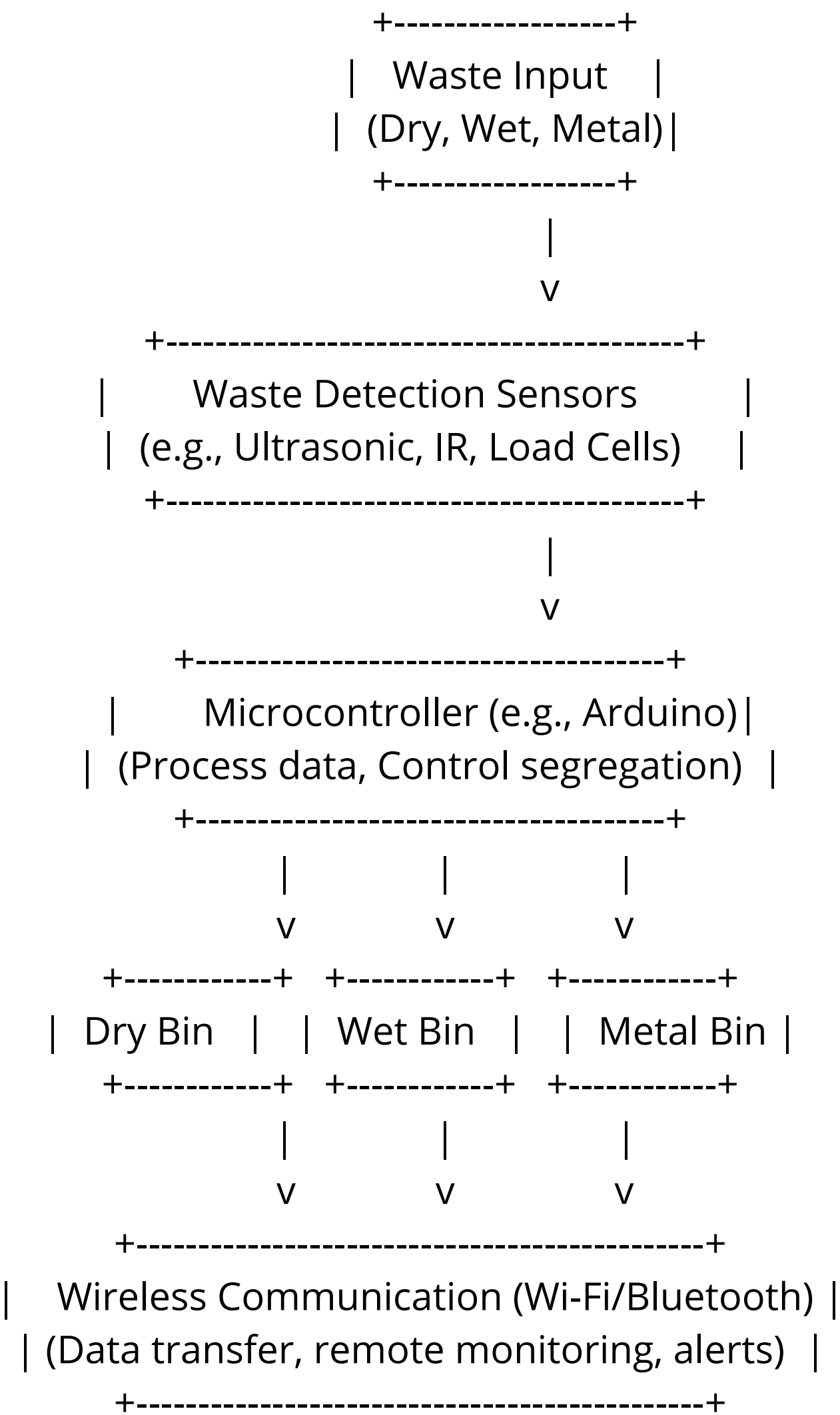
- When a dustbin is full, it will send a notification on the connected device that the trash in that dustbin needs to be cleared.

COST ESTIMATION

- **Estimated Cost: Around ₹1,760 - 2,000**

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|-------------------------------------|---------------------------------|
| • Arduino Board | : ₹400 - ₹800 |
| • Ultrasonic Sensor (HC-SR04) | : ₹100 - ₹200 |
| • IR Sensors (e.g., TCS3200) | : ₹50 - ₹150 (each, 2 needed) |
| • Metal Detection Sensor | : ₹300 - ₹700 |
| • Servo Motors (SG90) | : ₹150 - ₹300 (each, 3 needed) |
| • DC Motor (for lid opening) | : ₹150 - ₹300 |
| • Jumper Wires and Breadboard | : ₹150 - ₹300 |
| • LEDs (for indication) | : ₹10 - ₹50 (per LED, 3 needed) |
| • Buzzer | : ₹50 - ₹150 |
| • Power Supply (9V/12V Adapter) | : ₹100 - ₹300 |
| • Enclosure (Plastic/Cardboard Box) | : ₹300 - ₹600 |

BLOCK DIAGRAM





THANK YOU